

# Assignment 04 - Website Automation Agent

## Assignment Overview

Create an intelligent website automation agent capable of interacting with web pages autonomously. This project serves as a mini-version of browser automation tools like Browser Use, demonstrating your understanding of AI-driven browser control and web automation.

## Project Objective

Build an automation agent that can navigate to a specific URL, interact with web elements, and perform form-filling operations without manual intervention. Your agent should demonstrate intelligent decision-making in identifying and manipulating web page elements.

## Target Task

**Your automation agent must perform the following task:**

- Navigate to: <https://ui.shadcn.com/docs/forms/react-hook-form>
- Identify the form elements on the page (Name and Description fields)
- Automatically fill in the Name and Description

## Technical Requirements

### Required Tools & Technologies

**Browser Automation Library (Choose one):**

- Playwright (Recommended)
- Puppeteer

### Programming Language:

- Any language of your choice (JavaScript/TypeScript, Python, Java, etc.)

## Core Functionalities to Implement

**Your agent must include the following tools/capabilities:**

- **take\_screenshot** - Capture the current state of the browser window
- **open\_browser** - Initialize and launch a browser instance
- **navigate\_to\_url** - Direct the browser to a specific URL
- **click\_on\_screen(x, y)** - Perform mouse clicks at specified coordinates
- **send\_keys** - Input text into form fields or text areas
- **scroll** - Scroll the page to reveal hidden elements
- **double\_click** - Perform double-click actions when necessary

## Implementation Guidelines

- **Agent Architecture:** Design your agent with modular tools that can be composed together
- **Element Detection:** Implement intelligent element identification using selectors, XPath, or visual recognition
- **Error Handling:** Include proper error handling for network issues, element not found, and timeout scenarios
- **Logging:** Implement comprehensive logging to track agent actions and decisions
- **Configuration:** Use environment variables or configuration files for API keys and settings

## Project Deliverables

- **Source Code:** Complete, well-documented codebase with clear comments
- **README File:** Include setup instructions, dependencies, and how to run the agent
- **Architecture Document:** Brief explanation of your design decisions and agent workflow

## Evaluation Criteria

**Your project will be evaluated during your viva based on:**

- **Functionality (40%):** Does the agent successfully complete the target task?
- **Code Quality (25%):** Is the code clean, modular, and well-documented?
- **Agent Intelligence (20%):** How well does the agent handle element detection and decision-making?
- **Error Handling (10%):** Robustness in handling edge cases and failures
- **Documentation (5%):** Quality of README and inline code comments

## Submission Guidelines

- **Deadline:** No fixed deadline — evaluation will occur during your viva voce examination
- **Presentation:** Be prepared to demonstrate your agent live during the viva

## Helpful Resources

- **Playwright Documentation:** <https://playwright.dev/>
- **Puppeteer Documentation:** <https://pptr.dev/>
- **Browser Use (Reference):** <https://github.com/browser-use/browser-use>
- **OpenAI Agents SDK:** <https://openai.github.io/openai-agents-js>